

From Infinite Support to an Initial Full-RSB Interval

Readable reference edition with a novelty and dependency map

Based on the fully fixed manuscript and successive proof audits

July 14, 2026

What this edition changes

The original 23-page manuscript is retained unchanged after this guide. Several analytically important qualifications were typeset in compressed, small-font notes. This edition adds a normal-size reader's guide and an expanded Appendix R that restate those qualifications, the derivative bookkeeping, and the maximum-principle boundary data in readable form. The theorem and the proof in the main manuscript are not altered.

The result at a glance

Prior result used as the starting point

Lopatto proved that for every inverse temperature $\beta > 1$, the zero-field SK Parisi measure μ_β has infinite support.

Strengthening proved in the manuscript

For every $\beta > 1$, there is $q_\beta > 0$ such that

$$[0, q_\beta] \subseteq \text{supp } \mu_\beta.$$

Writing $u = u_{\mu_\beta}$ for the Parisi-PDE solution, the proof also gives

$$\mu_\beta(\{0\}) = 0, \quad \beta u_{xx}(0, 0) = 1, \quad u_{xxxx}(0, 0) < 0,$$

and the distribution function of μ_β is C^∞ on $[0, q_\beta)$.

The distinction is substantial: an infinite support may still be a highly disconnected set, whereas the new conclusion produces a genuine nondegenerate interval beginning at the origin.

Scope of the conclusion

The theorem does *not* claim that the entire support is a single interval, and it does not exclude gaps or atoms away from the initial interval. The manuscript remains an independent derivation that has not undergone external peer review; specialist review is appropriate because the conclusion is unusually strong.

Core proof chain

1. Extend the five slope-coordinate inequalities from finite atomic profiles to an arbitrary Parisi profile.
2. Prove a strict generalized first-gap inequality directly for the limiting profile.
3. Use Parisi optimality and stability to rule out an isolated origin.
4. Deduce exact marginality and show that the Parisi measure has no atom at zero.
5. Obtain $u_{xxxx}(0, 0) < 0$, compute Γ''' on a support gap, and exclude every sufficiently small gap by strict convexity and the Hermite–Hadamard inequality.

Established inputs versus new mathematical input

The manuscript deliberately separates literature inputs from the steps proved inside the note.

Established results used

Auffinger–Chen: Parisi-PDE regularity and profile continuity.

Evenness, strict convexity, boundedness and continuity of all positive-order spatial derivatives, together with uniform convergence of those derivatives when the order parameter converges. These results make it possible to pass slope-coordinate expressions and stochastic identities to a limiting profile.

Auffinger–Chen and Jagannath–Tobasco: optimizer and support conditions.

The facts that $0 \in \text{supp } \mu_\beta$, that support points minimize the first-variation function G , and that at every support point q one has

$$\Gamma(q) = q, \quad \Gamma'(q) = \beta^2 \mathbb{E}[u_{xx}(q, X_q)^2] \leq 1.$$

These are the variational equalities and stability inequalities used at the endpoints of a hypothetical gap.

Jagannath–Tobasco: weak-time regularity and approximation framework.

The time regularity needed to differentiate spatial derivatives on a constant-profile interval, together with atomic approximation. The manuscript supplies the flat-segment approximation in the exact form needed for the higher Itô identities.

Lopatto: finite-profile slope-coordinate mechanism.

The local slope-coordinate evolution equations, the five cone inequalities for finite Cole–Hopf compositions, and the infinite-support theorem. Appendix A reproduces the finite-profile preservation argument so the closure step can be checked independently.

Zhou: low-temperature FRSB context.

A prior interval result in a low-temperature regime, used as context for the meaning of full replica symmetry breaking rather than as an input to the all- $\beta > 1$ argument.

Genuinely new steps in this manuscript

Three central new inputs

1. **Uniform endpoint expansion for arbitrary profiles (Lemma 3.2).** The expansion through e^{-6x} has an $O(e^{-8x})$ remainder that may be differentiated to every required spatial order, uniformly along atomic approximations.
2. **Closure of the slope cone (Proposition 4.2).** All five slope inequalities pass from finite atomic profiles to an arbitrary Parisi profile, including the endpoint cases $c = 0$ and $c = 1$.
3. **Direct strict first-gap inequality (Proposition 5.1).** Strictness is proved anew for the limiting profile by a zero-crossing argument; it is not obtained by taking a limit of strict finite-profile inequalities, which could lose strictness.

The downstream new deductions are: accumulation of support at zero, exact marginality, absence of an atom at zero, a strict fourth-derivative sign, positivity of Γ''' on sufficiently small support gaps, and finally the exclusion of all such gaps.

Logical chain

finite-profile cone $\implies$ arbitrary-profile cone
 $\implies$ strict first-gap inequality $\implies [0, q_\beta] \subseteq \text{supp } \mu_\beta$.

Readability and revision map

The small-font passages in the original manuscript are not side comments; they contain the analytic repairs that make the proof self-contained relative to the quoted literature. Appendix R, placed after the main manuscript, restates them at normal size.

Main location	Normal-size restatement in Appendix R
Page 1	Corrected sign identity, revision history, theorem status and scope.
Page 5	Flat-segment atomic approximations and passage of the higher Itô identities.
Page 7	Boundary traces at $x = X$ and the weighted half-line estimate.
Pages 9–10	Endpoint cases in the closure argument; exact hypotheses for semigroup invariance and the generalized gap inequality.
Pages 12–14	Dominated differentiation, the upward zero-crossing property, and the one-crossing argument for q'' .
Pages 19–22	Direct fixed-flow endpoint expansion, derivative-order budget, endpoint ODE data, and the precise maximum-principle strips and lateral boundary values.

Sign-critical identity

$$\frac{w_B}{w} = -\frac{N+z}{C}.$$

All later identities in the zero-crossing proof use this corrected sign.

From Infinite Support to an Initial Full-RSB Interval

A corrected and analytically completed audit of the slope-coordinate extension

Independent mathematical audit

July 14, 2026

Abstract

Let μ_β be the zero-field Parisi measure of the Sherrington–Kirkpatrick model in the normalization $\xi(q) = \beta^2 q^2/2$. Lopatto recently proved that μ_β has infinite support for every $\beta > 1$. This note audits an extension of his slope-coordinate method and proves the stronger conclusion that, for every $\beta > 1$, there is $q_\beta > 0$ such that $[0, q_\beta] \subseteq \text{supp } \mu_\beta$. The main new input is a closure theorem: the five slope-coordinate inequalities, first obtained for finite compositions of Cole–Hopf transforms, pass to an arbitrary Parisi profile. The proof uses uniform convergence of all spatial derivatives of Parisi-PDE solutions and a profile-uniform endpoint expansion. The resulting generalized first-gap inequality rules out an isolated origin. Accumulation of support at zero then yields exact marginality $\beta u_{xx}(0, 0) = 1$ and excludes an atom at zero. The closed slope cone forces $u_{xxxx}(0, 0) < 0$, and strict convexity of the overlap derivative on every sufficiently small support gap excludes all such gaps.

Revision and analytic-completion note. The sign-critical identity is

$$\frac{w_B}{w} = -\frac{N+z}{C}.$$

The present revision incorporates the repairs from successive audit passes. Besides the corrected sign in the displayed identity, Lemma 2.3 now constructs flat-segment atomic approximations explicitly; Lemma 3.2 verifies the half-line boundary bounds used in Lemma 3.1; Lemma 4.4 and Proposition 5.1 state the precise smoothness, convexity, slope-range, bounded-derivative, and endpoint hypotheses; Proposition 4.2 covers the case $c = 1$; Proposition 5.1 refers explicitly to interior zeros; and Proposition A.3 records the domains and lateral boundary conditions in the maximum-principle steps. Most importantly, Lemma A.2 now proves the fixed-flow endpoint expansion directly from the initial expansion, rather than invoking Lemma 3.2 outside its stated class. No theorem statement or sign-sensitive algebraic identity is changed.

Status. This is an independent derivation of a new strengthening of [3]; it has not undergone external peer review. The formerly omitted analytic steps are supplied below, relative only to the standard Parisi-PDE regularity and optimizer conditions quoted from [1, 2]. Because the conclusion would settle the all-temperature initial-interval form of the SK full-RSB conjecture, specialist review remains essential.

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1 Statement and proof architecture

Fix $\beta > 0$. For a Borel probability measure ν on $[0, 1]$, write

$$\alpha_\nu(s) = \nu([0, s]).$$

Values of α_ν at individual jump times are immaterial for the PDE. Let u_ν denote the solution of

$$\partial_s u_\nu(s, x) + \frac{\beta^2}{2} (u_{\nu,xx}(s, x) + \alpha_\nu(s) u_{\nu,x}(s, x)^2) = 0, \quad u_\nu(1, x) = \log \cosh x. \quad (1)$$

The Parisi functional has a unique minimizer, denoted μ_β . We use the rigorous convention that the model is full replica-symmetry breaking (FRSB) when the support of its Parisi measure contains a nondegenerate interval, as in [4].

Theorem 1.1 (Initial FRSB interval). *For every $\beta > 1$, there is $q_\beta > 0$ such that*

$$[0, q_\beta] \subseteq \text{supp } \mu_\beta. \quad (2)$$

If $u = u_{\mu_\beta}$, then additionally

$$\mu_\beta(\{0\}) = 0, \quad \beta u_{xx}(0, 0) = 1, \quad u_{xxxx}(0, 0) < 0. \quad (3)$$

Consequently, the distribution function of μ_β is C^∞ on $[0, q_\beta)$.

The proof is modular.

- (i) Sections 2–4 show that the slope-coordinate inequalities of [3] hold for $u_\nu(q, \cdot)$ for an arbitrary profile above q .
- (ii) Section 5 proves a generalized first-gap inequality directly for the limiting profile; strictness is not obtained by taking a limit of strict finite-profile inequalities.
- (iii) Section 6 applies this inequality to show that 0 is an accumulation point of $\text{supp } \mu_\beta$, then proves exact marginality and absence of an atom at 0.
- (iv) Section 7 uses the slope cone at parameter zero to obtain $u_{xxxx}(0, 0) < 0$, computes Γ''' on a support gap, and excludes all sufficiently small gaps.

2 Parisi optimality and regularity

Throughout the rest of the note, unless otherwise stated,

$$\mu = \mu_\beta, \quad u = u_\mu, \quad \alpha = \alpha_\mu.$$

Let X solve the optimal-diffusion SDE

$$dX_s = \beta^2 \alpha(s) u_x(s, X_s) ds + \beta dW_s, \quad X_0 = 0, \quad (4)$$

and define

$$\Gamma(s) = \mathbb{E}[u_x(s, X_s)^2], \quad G(s) = \int_s^1 \frac{\beta^2}{2} (\Gamma(r) - r) dr. \quad (5)$$

Proposition 2.1 (Established Parisi facts). *Let ν be an arbitrary probability measure on $[0, 1]$.*

(a) *The function u_ν is even and strictly convex in x . Every spatial derivative $\partial_x^j u_\nu$, $j \geq 1$, is bounded and continuous on $[0, 1] \times \mathbb{R}$, with $|u_{\nu,x}| \leq 1$ and $u_{\nu,xx} > 0$. For each fixed j , the derivative bound can be chosen uniformly over all probability measures ν .*

(b) *If $\nu_n \rightarrow \nu$ in*

$$d(\nu_n, \nu) = \int_0^1 |\nu_n([0, s]) - \nu([0, s])| ds,$$

then for every $j \geq 0$,

$$\partial_x^j u_{\nu_n} \longrightarrow \partial_x^j u_\nu \quad \text{uniformly on } [0, 1] \times \mathbb{R}.$$

(c) *Every point of $\text{supp } \mu$ minimizes G , and $0 \in \text{supp } \mu$.*

(d) *For every $q \in \text{supp } \mu$,*

$$\Gamma(q) = q, \quad \beta^2 \mathbb{E}[u_{xx}(q, X_q)^2] \leq 1. \quad (6)$$

Proof. The regularity and convergence assertions are Propositions 1–2 of [1], specialized to $\xi(q) = \beta^2 q^2/2$. Uniformity over ν follows from compactness of the space of probability measures on $[0, 1]$ and the sup-norm continuity of each derivative map: otherwise a sequence with diverging derivative norm would have a weakly convergent subsequence, contradicting uniform derivative convergence. Part (c) combines Theorem 1 of [1] with the minimizer characterization in Corollary 3.6 of [2]. Part (d) is Proposition 1.1 of [2]. $\square$

Lemma 2.2 (Nontriviality below the critical temperature). *If $\beta > 1$, then $\mu_\beta \neq \delta_0$.*

Proof. For $\mu = \delta_0$, one has $\alpha(s) = 1$ for $s \geq 0$ and

$$u_\mu(s, x) = \frac{\beta^2}{2}(1 - s) + \log \cosh x.$$

Thus $u_{\mu,xx}(0, 0) = 1$. If δ_0 were minimizing, the stability inequality in (6) at $q = 0$ would give $\beta^2 \leq 1$, a contradiction. $\square$

Lemma 2.3 (Arbitrary-profile Itô identities). *Let ν be an arbitrary probability measure on $[0, 1]$, put $u = u_\nu$, and let X solve*

$$dX_s = \beta^2 \alpha_\nu(s) u_x(s, X_s) ds + \beta dW_s, \quad X_0 = 0.$$

Define

$$M_s = u_x(s, X_s), \quad C_s = u_{xx}(s, X_s), \quad D_s = u_{xxx}(s, X_s), \quad \Gamma_\nu(s) = \mathbb{E}[M_s^2].$$

Then, as identities of continuous semimartingales,

$$dM_s = \beta C_s dW_s, \quad dC_s = -\beta^2 \alpha_\nu(s) C_s^2 ds + \beta D_s dW_s. \quad (7)$$

Consequently,

$$\Gamma'_\nu(s) = \beta^2 \mathbb{E}[C_s^2] \quad (8)$$

for every s , and, in the absolutely continuous sense,

$$\Gamma'_\nu(s) - \Gamma'_\nu(0) = \beta^4 \int_0^s \mathbb{E}[D_r^2 - 2\alpha_\nu(r) C_r^3] dr. \quad (9)$$

On every open interval on which $\alpha_\nu \equiv m$ is constant, the analogous Itô identities for higher spatial derivatives hold on compact subintervals. In particular, with $R_s = u_{xxxx}(s, X_s)$,

$$dD_s = -3\beta^2 m C_s D_s ds + \beta R_s dW_s.$$

Proof. Choose finitely atomic measures ν_n such that

$$\int_0^1 |\alpha_{\nu_n}(s) - \alpha_\nu(s)| ds \rightarrow 0,$$

and write $\alpha_n = \alpha_{\nu_n}$ and $u_n = u_{\nu_n}$. Couple the corresponding diffusions X^n and X with the same Brownian motion. Proposition 2.1 gives uniform convergence of every spatial derivative of u_n to the corresponding derivative of u . In particular, $u_{n,x}$ are uniformly Lipschitz in x . Hence

$$\mathbb{E} \sup_{r \leq s} |X_r^n - X_r|^2 \leq C \left(\|u_{n,x} - u_x\|_\infty^2 + \|\alpha_n - \alpha_\nu\|_{L^1}^2 + \int_0^s \mathbb{E} \sup_{v \leq r} |X_v^n - X_v|^2 dr \right),$$

so Gronwall's inequality implies $X^n \rightarrow X$ in $L^2(\Omega; C[0, 1])$.

For a step profile, u_n is classically smooth between consecutive jump times. Applying Itô's formula on each such time slab and concatenating the identities is legitimate because u_n and all positive-order spatial derivatives are continuous across the jump times. Thus, with

$$M_s^n = u_{n,x}(s, X_s^n), \quad C_s^n = u_{n,xx}(s, X_s^n), \quad D_s^n = u_{n,xxx}(s, X_s^n),$$

one has

$$M_s^n - M_0^n = \beta \int_0^s C_r^n dW_r,$$

and

$$C_s^n - C_0^n = -\beta^2 \int_0^s \alpha_n(r) (C_r^n)^2 dr + \beta \int_0^s D_r^n dW_r.$$

Uniform convergence of the spatial derivatives, the convergence $X^n \rightarrow X$, boundedness of the derivatives, dominated convergence for the drift terms, and Itô's isometry for the stochastic terms allow passage to the limit. This proves (7) in integrated form for the arbitrary profile.

Applying Itô's formula to M_s^2 and taking expectations gives

$$\Gamma_\nu(s) - \Gamma_\nu(0) = \beta^2 \int_0^s \mathbb{E}[C_r^2] dr.$$

The integrand is continuous because u_{xx} and X are continuous and u_{xx} is bounded. This proves (8) for every s . Applying Itô's formula to C_s^2 gives

$$\mathbb{E}[C_s^2] - \mathbb{E}[C_0^2] = \beta^2 \int_0^s \mathbb{E}[D_r^2 - 2\alpha_\nu(r) C_r^3] dr,$$

which, together with (8), is (9).

Finally, let $I = [r_0, r_1]$ be compactly contained in an open interval on which $\alpha_\nu \equiv m$. Choose $a < r_0 < r_1 < b$ in that interval. Then $\nu((a, b]) = 0$, $\nu([0, a]) = m$, and $\nu((b, 1]) = 1 - m$. Approximate the two finite measures $\nu|_{[0, a]}$ and $\nu|_{(b, 1]}$ separately by finitely atomic measures with exactly the same respective total masses (allowing the zero measure when $m = 0$ or $m = 1$), and let ν_n be their sum. Then $\nu_n \rightarrow \nu$ in d and $\alpha_n \equiv m$ on $[a, b]$, hence on I . For each n , differentiating the PDE three times and applying Itô's formula along the optimal diffusion cancels the transport and fifth-derivative terms and gives

$$du_{n,xxx}(s, X_s^n) = -3\beta^2 m u_{n,xx}(s, X_s^n) u_{n,xxx}(s, X_s^n) ds + \beta u_{n,xxxx}(s, X_s^n) dW_s.$$

Uniform convergence of the required spatial derivatives, together with dominated convergence and Itô isometry, passes this identity to the limit on I . Repeating the differentiated calculation gives every higher identity used below. This agrees with the weak-time regularity $\partial_s \partial_x^2 u \in L^\infty$ in Proposition 8.1 of [2].

For $\nu = \mu$, one has $\Gamma_\nu = \Gamma$ as defined in (5). All stochastic integrals below are true martingales because the spatial derivatives are bounded. We shall use (8)–(9) in this minimizing-profile case without further comment.

3 Endpoint asymptotics for arbitrary profiles

The finite Cole–Hopf proof uses a three-term expansion as $x \rightarrow +\infty$. For a general Parisi profile, the uniform differentiated remainder requires justification. We first record a weighted parabolic estimate that makes this step explicit.

Lemma 3.1 (Weighted half-line estimate). *Fix $T, X, \lambda > 0$ and integers $L, M \geq 0$. Suppose R and its spatial derivatives through order L are continuous on $[0, T] \times [X, \infty)$, and that on each of finitely many time slabs R is a classical solution of*

$$R_r = \frac{1}{2} R_{xx} + b R_x + c R + F \quad \text{on } (0, T] \times [X, \infty), \quad (10)$$

where the x -derivatives of both b and c through order L are uniformly bounded. Assume that, uniformly for $0 \leq r \leq T$, R and its x -derivatives through order L grow at most like $1 + (x - X)^M$, and assume

$$|\partial_x^j F(r, x)| \leq A_j e^{-\lambda x}, \quad 0 \leq j \leq L,$$

that $\partial_x^j R(0, x) = O(e^{-\lambda x})$, and that the boundary values $\partial_x^j R(r, X)$ are uniformly bounded for $0 \leq j \leq L$. Then

$$|\partial_x^j R(r, x)| \leq C_j e^{-\lambda x}, \quad 0 \leq r \leq T, \quad x \geq X, \quad 0 \leq j \leq L. \quad (11)$$

The constants depend only on the displayed bounds and on M , not on the number of time intervals on which the coefficients are smooth.

Proof. Write

$$\mathcal{L} = \partial_r - \frac{1}{2} \partial_{xx} - b \partial_x - c.$$

For $j = 0$, let

$$H(r, x) = A e^{\gamma r} e^{-\lambda(x-X)}.$$

Choose

$$\gamma > \frac{\lambda^2}{2} + \lambda \|b\|_\infty + \|c_+\|_\infty + 1.$$

Then $\mathcal{L}H \geq H$. Taking A sufficiently large makes H dominate the initial data, the boundary data at $x = X$, and $|F|$.

The escape barrier must dominate the actual polynomial growth. Put

$$Q(x) = 1 + (x - X)^{2M+2}.$$

Because b and c_+ are bounded and $Q'/Q, Q''/Q$ are bounded on $[X, \infty)$, one may choose ρ so large that

$$\mathcal{L}(e^{\rho r} Q(x)) \geq e^{\rho r} Q(x).$$

For $\eta > 0$, apply the parabolic maximum principle on $[0, T] \times [X, Y]$ to

$$H + \eta e^{\rho r} Q \pm R.$$

Its parabolic boundary values are nonnegative once Y is sufficiently large, since Q has degree $2M + 2$ whereas R has growth degree at most M . Its $\mathcal{L}$ -image is nonnegative. Letting $Y \rightarrow \infty$ and then $\eta \downarrow 0$ proves (11) for $j = 0$.

For $R_j = \partial_x^j R$, differentiation of (10) yields

$$(R_j)_r = \frac{1}{2}(R_j)_{xx} + b(R_j)_x + c_j R_j + F_j,$$

where $c_j = c + j b_x$ is bounded, and F_j is a sum of $\partial_x^j F$ and bounded coefficient derivatives multiplying R_i with $i < j$. The hypotheses require precisely the derivatives of b and c through order j . Induction and the already proved estimates give $F_j = O(e^{-\lambda x})$, so the same barrier argument applies to R_j . If the coefficients have finitely many time jumps, the single global barrier works on every smooth time interval, and continuity of R_j propagates the comparison across the jumps. This proves the asserted uniformity. $\square$

Lemma 3.2 (Uniform endpoint expansion). *Let ν be any probability measure on $[0, 1]$, let $0 \leq q \leq 1$, and put $\psi(x) = u_\nu(q, x)$. Then, as $x \rightarrow +\infty$,*

$$\psi(x) = x + c_0 + d_1 e^{-2x} + d_2 e^{-4x} + d_3 e^{-6x} + O(e^{-8x}), \quad d_1 > 0, \quad (12)$$

with differentiated remainder of every order: for each $\ell \geq 0$,

$$\partial_x^\ell O(e^{-8x}) = O_\ell(e^{-8x}).$$

The constants may depend on ν, q, β , and ℓ .

Proof. Set $T = \beta^2(1 - q)$ and reverse time by

$$v(r, x) = u_\nu(1 - r/\beta^2, x), \quad 0 \leq r \leq T.$$

Then

$$v_r = \frac{1}{2}(v_{xx} + a(r)v_x^2), \quad v(0, x) = \log \cosh x, \quad 0 \leq a(r) \leq 1, \quad (13)$$

where $a(r) = \nu([0, 1 - r/\beta^2])$ up to a null set.

First suppose a is a step function. Let $y = e^{-2x}$ and set

$$P(r, x) = x + c(r) + d_1(r)y + d_2(r)y^2 + d_3(r)y^3. \quad (14)$$

Matching the coefficients of $1, y, y^2, y^3$ in (13) gives the absolutely continuous system

$$c' = \frac{a}{2}, \quad (15)$$

$$d'_1 = 2(1 - a)d_1, \quad (16)$$

$$d'_2 = (8 - 4a)d_2 + 2ad_1^2, \quad (17)$$

$$d'_3 = (18 - 6a)d_3 + 8ad_1d_2, \quad (18)$$

with

$$c(0) = -\log 2, \quad d_1(0) = 1, \quad d_2(0) = -\frac{1}{2}, \quad d_3(0) = \frac{1}{3}.$$

In particular,

$$d_1(r) = \exp\left(2 \int_0^r (1 - a(\rho)) d\rho\right) > 0. \quad (19)$$

Because $0 \leq a \leq 1$ and $r \leq T$, Gronwall's inequality gives profile-independent bounds for c, d_1, d_2, d_3 .

Let $R = v - P$ and

$$F = P_r - \frac{1}{2}(P_{xx} + aP_x^2).$$

The coefficient equations cancel all powers through y^3 , so for every ℓ ,

$$|\partial_x^\ell F(r, x)| \leq A_{\ell, T} e^{-8x} \quad (20)$$

for all sufficiently large x , uniformly in the step profile. Moreover,

$$R_r = \frac{1}{2}R_{xx} + bR_x - F, \quad b = \frac{a}{2}(v_x + P_x). \quad (21)$$

On a half-line $[X, \infty)$, every required x -derivative of b is bounded uniformly in the step profile: the derivatives of P are uniformly bounded there and Proposition 2.1(a) gives profile-independent bounds for the derivatives of v . Moreover, v and P have at most linear growth, while their positive-order spatial derivatives are bounded, so the polynomial-growth hypothesis of Lemma 3.1 holds with a fixed degree. The initial remainder and all of its derivatives are $O(e^{-8x})$ because

$$\log \cosh x = x - \log 2 + e^{-2x} - \frac{1}{2}e^{-4x} + \frac{1}{3}e^{-6x} + O(e^{-8x}).$$

At $x = X$, the Cole–Hopf representation gives $\sup_{0 \leq r \leq T} |v(r, X)| \leq |X| + C_{\beta, T}$; Proposition 2.1(a) bounds every positive spatial derivative of v , while P and its derivatives are uniformly bounded there. Hence all boundary values required by Lemma 3.1 are uniform. Applying that lemma inductively after differentiating (21) (where the zero-order coefficient is $c = 0$) gives

$$|\partial_x^\ell R(r, x)| \leq C_{\ell, T} e^{-8x}, \quad 0 \leq r \leq T, \quad (22)$$

with constants independent of the number and locations of the jumps of a .

For general ν , choose finitely atomic $\nu_n \rightarrow \nu$ in the metric of Proposition 2.1. Their reversed profiles a_n converge to a in $L^1[0, T]$. The finite-dimensional systems (15)–(18) and Gronwall's inequality imply uniform convergence of the corresponding coefficients. Proposition 2.1(b) gives uniform convergence of every spatial derivative of the PDE solutions. Passing to the limit in (22) proves (12) with all differentiated remainders, and (19) gives $d_1 > 0$. $\square$

Corollary 3.3 (Slope-endpoint expansion). *Let $B = \psi'(x)$ and $C(B) = \psi''(x(B))$. With $\delta = 1 - B$,*

$$C(B) = 2\delta + c_2\delta^2 + c_3\delta^3 + O(\delta^4), \quad (23)$$

where the remainder may be differentiated three times, losing one power of δ at each derivative. In particular,

$$x(B) = \frac{1}{2} \log \frac{1}{1-B} + O(1), \quad C(B) = 2(1-B) + O((1-B)^2). \quad (24)$$

Proof. With $y = e^{-2x}$, Lemma 3.2 gives

$$1 - B = 2d_1y + 4d_2y^2 + 6d_3y^3 + O(y^4), \quad C = 4d_1y + 16d_2y^2 + 36d_3y^3 + O(y^4).$$

Since $d_1 > 0$, the first relation has a smooth inverse

$$y = b_1\delta + b_2\delta^2 + b_3\delta^3 + O(\delta^4)$$

with the corresponding differentiated bounds. Substitution into the second relation yields (23); the leading coefficient is $4d_1/(2d_1) = 2$. The first relation also gives (24). $\square$

4 The closed slope cone

Let ψ be even and strictly convex, with $\psi'(\mathbb{R}) = (-1, 1)$. For $B = \psi'(x)$, write $x = x(B)$ and

$$C(B) = \psi''(x(B)).$$

Fix a parameter $a \geq 0$. For $B \neq 0$, define

$$K_a(B) = -\frac{C_B(B)}{2B} - a, \quad J_a(B) = -a - \frac{1}{2}C_{BB}(B). \quad (25)$$

Evenness makes both definitions smooth at $B = 0$, and

$$C_B = -2(a + K_a)B, \quad J_a = K_a + B(K_a)_B = (BK_a)_B. \quad (26)$$

Definition 4.1 (Closed slope cone). *We say that (ψ, a) belongs to the closed slope cone if, for $0 \leq B < 1$,*

$$K_a \geq 0, \quad (K_a)_B \geq 0, \quad J_a \geq 0, \quad (J_a)_B \geq 0, \quad (C^{3/2}J_a)_B \leq 0. \quad (27)$$

Writing

$$z = -\frac{1}{2}C_B = (a + K_a)B, \quad (28)$$

the last inequality is equivalent to

$$C(J_a)_B \leq 3zJ_a. \quad (29)$$

The finite-profile cone is proved from the local evolution equations in Appendix A; see Corollary A.4. We now close it under arbitrary-profile limits.

Proposition 4.2 (Arbitrary-profile closure). *Let ν be any probability measure on $[0, 1]$, fix $q \in [0, 1]$, and set*

$$\psi(x) = u_\nu(q, x), \quad c = \nu([0, q]).$$

Then (ψ, a) belongs to the closed slope cone for every $a \in [0, c]$.

Proof. Only the profile on $(q, 1]$ affects the value $u_\nu(q, \cdot)$. Replace the part of ν below q by a single atom and define

$$\bar{\nu} = c\delta_q + \nu|_{(q,1]}. \quad (30)$$

Then $u_{\bar{\nu}}(q, \cdot) = u_\nu(q, \cdot) = \psi$. Assume first that $c > 0$. If $c = 1$, take $\lambda_n = 0$. If $0 < c < 1$, choose finitely atomic measures λ_n on $(q, 1]$, each of total mass $1 - c$, converging weakly to $\nu|_{(q,1]}$. In either case set

$$\nu_n = c\delta_q + \lambda_n, \quad \psi_n = u_{\nu_n}(q, \cdot).$$

Then $\nu_n \rightarrow \bar{\nu}$ in the metric d . Since ν_n is finitely atomic, ψ_n is obtained from log cosh by finitely many fixed-parameter Cole–Hopf evolutions and parameter decreases, ending at parameter c . Corollary A.4 gives (27) for (ψ_n, c) .

Keeping C_n fixed and decreasing the parameter from c to $a \leq c$ gives the exact identities

$$K_{n,a} = K_{n,c} + c - a, \quad J_{n,a} = J_{n,c} + c - a, \quad (J_{n,a})_B = (J_{n,c})_B, \quad (31)$$

and

$$(C_n^{3/2} J_{n,a})_B = (C_n^{3/2} J_{n,c})_B + \frac{3}{2}(c - a)C_n^{1/2}(C_n)_B. \quad (32)$$

The cone at parameter c implies

$$(C_n)_B = -2(c + K_{n,c})B \leq 0 \quad (B \geq 0),$$

so all five inequalities remain valid at parameter a .

We pass to the limit. Proposition 2.1(b) gives uniform convergence of all spatial derivatives of ψ_n to those of ψ . Fix a compact interval $I \Subset (-1, 1)$. Since ψ' is strictly increasing, the inverse maps

$$x_n(B) = (\psi'_n)^{-1}(B)$$

converge uniformly to $x(B) = (\psi')^{-1}(B)$ on I . Moreover, ψ'' is bounded below on the compact set $x(I)$, and the same holds for ψ''_n for all large n . The chain rule gives

$$(C_n)_B = \frac{\psi_n^{(3)}}{\psi_n''} \circ x_n, \quad (C_n)_{BB} = \frac{\psi_n^{(4)}\psi_n'' - (\psi_n^{(3)})^2}{(\psi_n'')^3} \circ x_n, \quad (33)$$

and $(C_n)_{BBB}$ is a continuous rational expression in

$$\psi_n'', \psi_n^{(3)}, \psi_n^{(4)}, \psi_n^{(5)}$$

evaluated at x_n . Thus C_n and its first three B -derivatives converge uniformly on I .

There is no singularity at $B = 0$. Since $C_B(0) = 0$,

$$\frac{C_B(B)}{B} = \int_0^1 C_{BB}(\tau B) d\tau, \quad (34)$$

and

$$\frac{BC_{BB}(B) - C_B(B)}{B^2} = \frac{1}{B^2} \int_0^B \int_v^B C_{BBB}(w) dw dv. \quad (35)$$

The same formulas hold for C_n , so convergence is uniform through $B = 0$. It follows that

$$K_{n,a}, (K_{n,a})_B, J_{n,a}, (J_{n,a})_B, (C_n^{3/2} J_{n,a})_B$$

converge uniformly on I . The inequalities pass to the limit on every such compact interval, hence on $0 \leq B < 1$.

If $c = 0$, choose $\varepsilon_n \downarrow 0$ and finitely atomic probability measures λ_n supported on $(q, 1]$ with $\lambda_n \rightarrow \nu$, and put

$$\nu_n = \varepsilon_n \delta_q + (1 - \varepsilon_n) \lambda_n.$$

The finite-profile cone holds at parameter ε_n . Formulas (31)–(32), with the decreased parameter equal to 0, preserve all signs. The same compact-slope convergence proves the result for $(\psi, 0)$. $\square$

Audit point 4.3 (Why strictness is not lost). *Proposition 4.2 passes only closed inequalities to the limit. The strict generalized gap inequality below is not obtained by taking a limit of strict finite-profile inequalities; such a limit could become an equality. Instead, Section 5 repeats the zero-crossing argument directly for the limiting profile.*

For $a > 0$ and $r \geq 0$, define the Cole–Hopf semigroup

$$T_{a,r}\psi(x) = \frac{1}{a} \log \mathbb{E} \exp(a\psi(x + \sqrt{r}Z)), \quad Z \sim N(0, 1). \quad (36)$$

Lemma 4.4 (Semigroup invariance). Suppose (ψ, a) belongs to the closed slope cone, $a > 0$, and suppose $\psi \in C^\infty(\mathbb{R})$, $\psi'(\mathbb{R}) = (-1, 1)$, $\psi'' > 0$, every positive-order spatial derivative of ψ is bounded, and ψ has the differentiated endpoint expansion (12). Then $(T_{a,r}\psi, a)$ belongs to the closed slope cone for every $r \geq 0$. The endpoint expansion and all derivatives needed in slope coordinates are uniform for r in a compact interval.

Proof. This is Proposition A.3. The Cole–Hopf formula also yields, under the associated tilted Gaussian law $Q_{r,x}$,

$$(T_{a,r}\psi)_{xx}(x) = \mathbb{E}_{Q_{r,x}}[\psi''(x + \sqrt{r}Z)] + a \operatorname{Var}_{Q_{r,x}}(\psi'(x + \sqrt{r}Z)) > 0,$$

so strict convexity and the slope coordinate persist. The remaining inputs in Appendix A are precisely the listed boundedness and endpoint hypotheses. $\square$

5 A generalized first-gap inequality

Let $m, t > 0$, and let ψ satisfy the same evenness, smoothness, strict-convexity, slope-range, bounded-derivative, and differentiated endpoint hypotheses as in Lemma 4.4, with (ψ, m) in the closed slope cone. Set

$$U(s, x) = T_{m,t-s}\psi(x), \quad 0 \leq s \leq t. \quad (37)$$

Let X solve

$$dX_s = mU_x(s, X_s) ds + d\widetilde{W}_s, \quad X_0 = 0, \quad (38)$$

and define

$$q(s) = \mathbb{E}[U_x(s, X_s)^2], \quad \theta = U_{xx}(0, 0) > 0. \quad (39)$$

Proposition 5.1 (Generalized gap inequality). *Under these hypotheses,*

$$2 \int_0^t q(s) ds = tq(t) \quad (40)$$

implies

$$q(t) < t\theta^2. \quad (41)$$

5.1 Brownian identities

Since

$$U_s = -\frac{1}{2}(U_{xx} + mU_x^2),$$

Itô's formula, with

$$M_s = U_x(s, X_s), \quad C_s = U_{xx}(s, X_s),$$

gives

$$dM_s = C_s d\widetilde{W}_s, \quad dC_s = -mC_s^2 ds + U_{xxx}(s, X_s) d\widetilde{W}_s. \quad (42)$$

Therefore

$$q(0) = 0, \quad q'(s) = \mathbb{E}[C_s^2], \quad q'(0) = \theta^2, \quad (43)$$

and

$$q''(s) = \mathbb{E}[U_{xxx}(s, X_s)^2 - 2mC_s^3], \quad q''(0) = -2m\theta^3 < 0. \quad (44)$$

Bounded spatial derivatives imply

$$q \in C^2([0, t]) \cap C^3((0, t)).$$

The one-time density of X_s can be identified by Girsanov. Under a reference Brownian motion W , define

$$L_s = \frac{\exp(mU(s, W_s))}{\exp(mU(0, 0))}.$$

The backward PDE makes L the stochastic exponential of $mU_x(s, W_s)$. Under $d\mathbb{Q} = L_t d\mathbb{P}$, the coordinate process satisfies (38). Hence, for $s > 0$, X_s has the even positive density

$$p_s(x) = \frac{1}{Z\sqrt{2\pi s}} \exp\left(mU(s, x) - \frac{x^2}{2s}\right), \quad Z = e^{mU(0,0)}. \quad (45)$$

5.2 The zero-crossing lemma

Lemma 5.2 (A zero of q'' is crossed upward). *For every $s \in (0, t)$,*

$$q''(s) = 0 \implies q'''(s) > 0. \quad (46)$$

Proof. Fix $s \in (0, t)$. In slope coordinates write

$$B = U_x(s, x), \quad C = U_{xx}(s, x), \quad x = x(s, B), \quad 0 < B < 1.$$

All slope quantities below use parameter m . To avoid the ambiguous product/derivative notation, set

$$P(B) = BK(B), \quad z = -\frac{1}{2}C_B = (m + K)B, \quad N = \frac{x}{s} + P, \quad \Phi = 2z^2 - mC. \quad (47)$$

Thus $P_B = J$. By Proposition 4.2 and Lemma 4.4,

$$K, K_B, J, J_B \geq 0, \quad CJ_B \leq 3zJ. \quad (48)$$

Define the positive weight

$$w(s, B) = 2p_s(x(s, B))C(s, B). \quad (49)$$

Since $U_{xxx} = CC_B = -2Cz$, changing variables by $dB = C dx$ and using evenness gives

$$\frac{1}{2}q''(s) = \int_0^1 w(s, B)\Phi(s, B) dB. \quad (50)$$

At fixed s , $x_B = C^{-1}$, $P_B = J$, and $z_B = m + J$, so

$$N_B = \frac{1}{sC} + J. \quad (51)$$

The sign-critical logarithmic derivative is obtained directly from (45) and (49):

$$\begin{aligned} \frac{w_B}{w} &= \frac{mB - x/s}{C} + \frac{C_B}{C} \\ &= \frac{mB - x/s - 2z}{C} = -\frac{N + z}{C}. \end{aligned} \quad (52)$$

Indeed, $z = mB + P$. In addition,

$$\Phi_B = 2z(2J + 3m) > 0 \quad (0 < B < 1), \quad (53)$$

because $z \geq mB > 0$.

The forward Cole–Hopf time is $r = t - s$. Proposition A.3 gives, at fixed B ,

$$x_s = -P, \quad C_s = C^2 J, \quad z_s = 2CzJ - \frac{1}{2}C^2 J_B. \quad (54)$$

Consequently,

$$\Phi_s = CJ\Phi + G, \quad G = 2Cz(3zJ - CJ_B) \geq 0. \quad (55)$$

Differentiating (49) at fixed B , using (45) and (54), gives

$$(\log w)_s = -\frac{1}{2s} + CJ - \frac{m}{2}C + PN - \frac{1}{2}z^2 - \frac{1}{2}P^2 + \frac{x^2}{2s^2}. \quad (56)$$

Set

$$R_1 = P\frac{x}{s} + \frac{1}{2}P^2, \quad R_0 = 2CJ - \frac{m}{2}C - \frac{1}{2}z^2. \quad (57)$$

If $q''(s) = 0$, then (50), (55), and (56) yield

$$\frac{d}{ds} \int_0^1 w\Phi dB = \int_0^1 w \left(\frac{x^2}{2s^2} \Phi + R_1 \Phi + R_0 \Phi + G \right) dB. \quad (58)$$

The differentiation is justified as follows. Uniformly for s in a compact subinterval of $(0, t)$, Corollary 3.3 and Lemma A.2 give

$$x(B) = \frac{1}{2} \log \frac{1}{1-B} + O(1), \quad C(B) = 2(1-B) + O((1-B)^2),$$

and K, J, z, R_0, G remain bounded as $B \uparrow 1$. Since U has linear growth, (45) gives a Gaussian bound in x , which dominates $|\partial_s(w\Phi)|$ by an integrable multiple of $w(1+x^2)$.

We estimate the terms in (58). If $W_0 = \int_0^1 w dB$ and R is nondecreasing, then, whenever $\int_0^1 w\Phi dB = 0$,

$$\int_0^1 w\Phi R dB = \frac{1}{2W_0} \int_0^1 \int_0^1 w(B)w(D)[\Phi(B) - \Phi(D)][R(B) - R(D)] dB dD. \quad (59)$$

Because both Φ and x^2 are strictly increasing,

$$\int_0^1 w\Phi x^2 dB > 0. \quad (60)$$

Moreover,

$$(R_1)_B = J\frac{x}{s} + \frac{P}{sC} + PJ \geq 0. \quad (61)$$

For the unbounded R_1 , apply (59) first to $R_1 \wedge M$. The Gaussian endpoint bound gives

$$w|\Phi|(1 + R_1) \in L^1(0, 1),$$

so dominated convergence yields

$$\int_0^1 wR_1\Phi dB \geq 0. \quad (62)$$

It remains to control $R_0\Phi + G$. Define

$$F(B) = \int_B^1 w(r)\Phi(r) dr, \quad \tau(B) = \frac{F(B)}{w(B)}. \quad (63)$$

Because Φ is strictly increasing and has zero w -mean, $F \geq 0$, hence $\tau \geq 0$. We claim

$$0 \leq \tau \leq Cz. \quad (64)$$

Set

$$\mathcal{H} = wCz - F, \quad L = CJ - z(N + z). \quad (65)$$

Using (52) and $(Cz)_B = CJ - \Phi$ gives exactly

$$\mathcal{H}_B = wL. \quad (66)$$

Further,

$$(CJ)_B = -2zJ + CJ_B \leq zJ, \quad (67)$$

whereas

$$[z(N + z)]_B = (m + J)(N + z) + z\left(\frac{1}{sC} + J + m + J\right) > zJ. \quad (68)$$

Thus $L_B < 0$. At $B = 0$, $z = 0$ and $F(0) = 0$, so $\mathcal{H}(0) = 0$. At $B = 1$, $F(1) = 0$ and the Gaussian endpoint bound gives $wCz \rightarrow 0$, so $\mathcal{H}(1) = 0$. Also $wL \in L^1(0, 1)$. Hence $\int_0^1 wL dB = 0$. Since L is strictly decreasing, it has one zero; integrating (66) from the nearer endpoint shows $\mathcal{H} \geq 0$. This is precisely $F \leq wCz$, proving (64).

The same endpoint bounds justify integration by parts. Since $F' = -w\Phi$ and $F(0) = F(1) = 0$,

$$\int_0^1 w\Phi R_0 dB = \int_0^1 w\tau(R_0)_B dB. \quad (69)$$

A direct differentiation gives

$$(R_0)_B = 2CJ_B - 5zJ. \quad (70)$$

Combining (55), (69), and (70),

$$\int_0^1 w(R_0\Phi + G) dB = \int_0^1 w[2CJ_B(\tau - Cz) + zJ(6Cz - 5\tau)] dB. \quad (71)$$

Because $\tau - Cz \leq 0$, multiplying $CJ_B \leq 3zJ$ by $2(\tau - Cz)$ reverses the inequality. Therefore

$$\int_0^1 w(R_0\Phi + G) dB \geq \int_0^1 wzJ\tau dB \geq 0. \quad (72)$$

Equations (58), (60), (62), and (72) imply

$$\int_0^1 w\Phi dB = 0 \implies \frac{d}{ds} \int_0^1 w\Phi dB > 0.$$

By (50), this is (46). □

5.3 Completion of the generalized gap inequality

Proof of Proposition 5.1. Integrating by parts twice and using $q(0) = 0$ gives

$$tq(t) - 2 \int_0^t q(s) ds = \int_0^t s(t-s)q''(s) ds. \quad (73)$$

Thus (40) is equivalent to the vanishing of the right-hand side. Put $g = q''$. By (44), $g(0) < 0$. Lemma 5.2 says that every zero of g in $(0, t)$ is crossed with positive derivative. Hence g has at most one zero in $(0, t)$. Indeed, after a zero a one has $g > 0$ immediately to its right; if there were a later zero, let $b > a$ be the first return. Then $g > 0$ on (a, b) and the left derivative of g at b would be nonpositive, contradicting $g'(b) > 0$. Since the weighted integral in (73) is zero while $g < 0$ near 0, the function g must be positive somewhere. Hence there is a unique $c \in (0, t)$ with

$$g < 0 \text{ on } (0, c), \quad g > 0 \text{ on } (c, t).$$

Define positive measures

$$d\eta_-(s) = (t-s)(-g(s))\mathbf{1}_{(0,c)}(s) ds, \quad d\eta_+(s) = (t-s)g(s)\mathbf{1}_{(c,t)}(s) ds.$$

The balance relation gives

$$\int s d\eta_- = \int s d\eta_+.$$

Since $s < c$ on the support of η_- and $s > c$ on the support of η_+ , strictly on sets of positive measure,

$$c\eta_+((0, t)) < \int s d\eta_+ = \int s d\eta_- < c\eta_-((0, t)).$$

Thus $\eta_+((0, t)) < \eta_-((0, t))$. Finally,

$$q(t) - tq'(0) = \int_0^t (t-s)q''(s) ds = \eta_+((0, t)) - \eta_-((0, t)) < 0.$$

Using $q'(0) = \theta^2$ proves (41). □

6 The origin: accumulation, marginality, and no atom

Proposition 6.1 (The origin is not isolated). *For $\beta > 1$, the point 0 is an accumulation point of $\text{supp } \mu_\beta$.*

Proof. Assume that 0 is isolated. Since $0 \in \text{supp } \mu$ and $\mu \neq \delta_0$ by Lemma 2.2, there is a first positive support point

$$q_* = \min(\text{supp } \mu \setminus \{0\}) > 0.$$

Isolation implies

$$m = \mu(\{0\}) > 0, \quad \alpha(s) = m \quad (0 \leq s < q_*).$$

Put

$$t = \beta^2 q_*, \quad \psi(x) = u(q_*, x), \quad U(r, x) = u(r/\beta^2, x) = T_{m, t-r} \psi(x). \quad (74)$$

Let

$$c = \alpha(q_*) = \mu([0, q_*]) \geq m.$$

Proposition 4.2 gives the closed slope cone for (ψ, c) , and the parameter-decrease formulas (31)–(32) give it for (ψ, m) . Hence Proposition 5.1 applies.

Under the time change $r = \beta^2 s$, the SDE (4) on $[0, q_*]$ becomes (38), and the function in (39) is

$$q(r) = \Gamma(r/\beta^2). \quad (75)$$

Both 0 and q_* minimize G , so

$$0 = G(0) - G(q_*) = \frac{\beta^2}{2} \int_0^{q_*} (\Gamma(s) - s) ds. \quad (76)$$

Also, $q_* \in \text{supp } \mu$, so $\Gamma(q_*) = q_*$. Rescaling (76) gives

$$2 \int_0^t q(r) dr = tq(t).$$

Proposition 5.1 therefore yields

$$q_* = q(t) < tu_{xx}(0, 0)^2 = \beta^2 q_* u_{xx}(0, 0)^2.$$

Thus $\beta^2 u_{xx}(0, 0)^2 > 1$, contradicting the stability condition (6) at the support point 0. $\square$

Proposition 6.2 (Exact marginality). *For $\beta > 1$,*

$$u_{xx}(0, 0) = \frac{1}{\beta}. \quad (77)$$

Proof. By Proposition 6.1, choose $q_n \in \text{supp } \mu$ with $q_n \downarrow 0$. Evenness gives $\Gamma(0) = 0$, and self-consistency gives $\Gamma(q_n) = q_n$. Since (8) makes Γ differentiable at zero,

$$\Gamma'(0) = \lim_{n \rightarrow \infty} \frac{\Gamma(q_n) - \Gamma(0)}{q_n} = 1.$$

Equation (8) at zero gives

$$\Gamma'(0) = \beta^2 u_{xx}(0, 0)^2.$$

Strict convexity gives $u_{xx}(0, 0) > 0$, so the positive square root must be chosen. $\square$

Proposition 6.3 (No atom at the origin). *For $\beta > 1$,*

$$\mu(\{0\}) = 0. \quad (78)$$

Proof. Let $m_0 = \mu(\{0\})$. Then $\alpha(s) \rightarrow m_0$ as $s \downarrow 0$. By continuity and boundedness of the spatial derivatives, $X_s \rightarrow 0$ in probability and in every finite moment, and therefore

$$C_s \longrightarrow u_{xx}(0,0) = \frac{1}{\beta}, \quad D_s \longrightarrow u_{xxx}(0,0) = 0$$

in every finite L^p . Dividing (9) by s and using dominated convergence gives the right derivative

$$\Gamma''(0+) = \beta^4 \left(0 - 2m_0 \frac{1}{\beta^3} \right) = -2\beta m_0. \quad (79)$$

Together with $\Gamma(0) = 0$ and $\Gamma'(0) = 1$, this gives

$$\Gamma(s) = s - \beta m_0 s^2 + o(s^2). \quad (80)$$

If $m_0 > 0$, then $\Gamma(s) < s$ for all sufficiently small $s > 0$, contradicting the sequence $q_n \downarrow 0$ of support points for which $\Gamma(q_n) = q_n$. Hence $m_0 = 0$. $\square$

7 Negative fourth derivative and exclusion of small gaps

7.1 A nondegenerate fourth derivative at the origin

Apply Proposition 4.2 to $\psi(x) = u(0, x)$. By Proposition 6.3, its endpoint parameter is $a = \alpha(0) = 0$. Let

$$B = u_x(0, x), \quad C(B) = u_{xx}(0, x(B)).$$

Proposition 7.1 (Strict fourth-order sign). *One has*

$$u_{xxxx}(0,0) < 0. \quad (81)$$

Proof. The closed slope cone gives $J \geq 0$ and

$$E(B) = C(B)^{3/2} J(B) \geq 0, \quad E_B \leq 0.$$

Suppose $J(0) = 0$. Then $E(0) = 0$, and monotonicity together with nonnegativity forces $E \equiv 0$ on $[0, 1)$, hence $J \equiv 0$. Since $J = (BK)_B$ and $BK(0) = 0$, one obtains $K \equiv 0$. Equation (26) at $a = 0$ then gives $C_B = 0$, so C is constant. This is impossible: Proposition 6.2 gives $C(0) = 1/\beta > 0$, whereas Corollary 3.3 gives $C(B) \rightarrow 0$ as $B \uparrow 1$. Therefore $J(0) > 0$.

At $x = 0$, one has $B = 0$ and $C_B(0) = 0$ by evenness. Since $\partial_x = C\partial_B$,

$$u_{xxx} = C(C_B)^2 + C^2 C_{BB}. \quad (82)$$

At $B = 0$, the first term vanishes and $C_{BB}(0) = -2J(0)$ because $a = 0$. Thus

$$u_{xxx}(0,0) = -2C(0)^2 J(0) < 0.$$

$\square$

7.2 The third derivative of the overlap function on a support gap

Let (a, b) be a connected component of $[0, 1] \setminus \text{supp } \mu$. Then $\alpha(s) = m$ is constant on (a, b) . Along the optimal diffusion define

$$R_s = u_{xxxx}(s, X_s).$$

On every compact subinterval of (a, b) , Lemma 2.3 permits differentiation of the Parisi PDE and application of Itô's formula to the resulting spatial derivatives. This gives

$$dC_s = -\beta^2 m C_s^2 ds + \beta D_s dW_s, \quad dD_s = -3\beta^2 m C_s D_s ds + \beta R_s dW_s. \quad (83)$$

Lemma 7.2 (Gap formula). *For $s \in (a, b)$,*

$$\Gamma'''(s) = \beta^6 \mathbb{E} \left[R_s^2 - 12m C_s D_s^2 + 6m^2 C_s^4 \right]. \quad (84)$$

Proof. Equation (9) becomes, on the gap,

$$\Gamma''(s) = \beta^4 \mathbb{E} [D_s^2 - 2m C_s^3].$$

From (83), Itô's formula gives

$$\frac{d}{ds} \mathbb{E} [D_s^2] = \beta^2 \mathbb{E} [R_s^2 - 6m C_s D_s^2],$$

and

$$\frac{d}{ds} \mathbb{E} [C_s^3] = \beta^2 \mathbb{E} [-3m C_s^4 + 3C_s D_s^2].$$

Substitution gives (84). The spatial derivatives are bounded, so the stochastic integrals are true martingales and differentiation under expectation is justified. The identities hold first on compact subintervals of the gap and hence at every point of (a, b) . $\square$

Proposition 7.3 (Positive third derivative on small gaps). *There exists $\varepsilon_\beta > 0$ such that, at every $s \in (0, \varepsilon_\beta)$ lying in a support gap,*

$$\Gamma'''(s) > 0. \quad (85)$$

Proof. Let $\kappa = u_{xxxx}(0, 0)$, which is nonzero by Proposition 7.1. As $s \downarrow 0$, Proposition 6.3 gives $\alpha(s) \rightarrow 0$, while continuity and boundedness of spatial derivatives and $X_s \rightarrow 0$ yield

$$C_s \rightarrow \frac{1}{\beta}, \quad D_s \rightarrow 0, \quad R_s \rightarrow \kappa$$

in every finite L^p . At a point s in a support gap the gap parameter is $m = \alpha(s)$. Therefore the expectation in brackets in (84) converges to $\kappa^2 > 0$ as $s \downarrow 0$ through such points. Choose $\varepsilon_\beta > 0$ so that this expectation is positive for every such $s \in (0, \varepsilon_\beta)$. $\square$

7.3 No small complementary interval

Proposition 7.4 (Initial interval in the support). *There exists $q_\beta > 0$ such that*

$$[0, q_\beta] \subseteq \text{supp } \mu. \quad (86)$$

Proof. By Proposition 6.1, choose

$$q_\beta \in \text{supp } \mu \cap (0, \varepsilon_\beta),$$

where ε_β is from Proposition 7.3. Suppose $[0, q_\beta]$ is not contained in the support. Then the open complement contains a connected component

$$(a, b) \subset (0, q_\beta).$$

If $a = 0$, then 0 would be isolated in the support, contradicting Proposition 6.1; hence $a > 0$. Closedness of the support gives $a, b \in \text{supp } \mu$.

On (a, b) , Proposition 7.3 says that Γ' is strictly convex. The strict Hermite–Hadamard inequality gives

$$\frac{1}{b-a} \int_a^b \Gamma'(s) \, ds < \frac{\Gamma'(a) + \Gamma'(b)}{2}. \quad (87)$$

The support equalities in (6) imply

$$\frac{1}{b-a} \int_a^b \Gamma'(s) \, ds = \frac{\Gamma(b) - \Gamma(a)}{b-a} = 1.$$

On the other hand, (8) and the stability inequalities in (6) give $\Gamma'(a), \Gamma'(b) \leq 1$. Thus (87) reads $1 < 1$, a contradiction. $\square$

Proof of Theorem 1.1. Proposition 7.4 proves (2). Propositions 6.2, 6.3, and 7.1 prove (3). Finally, Theorem 2(ii) of [1] states that when an interval lies in the support, the distribution function of the Parisi measure is infinitely differentiable up to the right endpoint from the left. This gives the last assertion. $\square$

8 Dependency audit, corrections, and scope

The proof uses the following external inputs.

Input	Role
Uniform convergence of all spatial derivatives of the Parisi PDE	Passes the five slope inequalities from atomic profiles to an arbitrary profile and supplies continuity and boundedness in the stochastic identities.
Weak-time regularity and atomic-profile approximation	Justifies the arbitrary-profile Itô identities, the absolutely continuous identity for Γ'' , and the higher differentiated identities on a support gap.
Support optimality and self-consistency	Gives $G(0) = G(q_*)$, $\Gamma(q) = q$, and $\Gamma'(q) \leq 1$ at support points.
Smoothness on support intervals	Converts the support conclusion into the usual smooth FRSB description near the origin.

The finite-profile slope cone is not left as a black box: Appendix A records the evolution equations, endpoint ODEs, degenerate maximum principle, base case $(\log \cosh, 1)$, and parameter-decrease step. The calculation is the one developed in [3], but the argument needed here is reproduced so that the arbitrary-profile closure can be checked independently.

The present revision incorporates every repair identified in successive audit passes. Lemma 3.1 uses an escape barrier whose degree dominates the assumed polynomial growth and requires exactly the coefficient derivatives produced by differentiation. Lemma 2.3 supplies the arbitrary-profile Itô passage and explicitly constructs approximants that remain flat on a prescribed constant-profile interval. Lemma 3.2 verifies the boundary bounds needed in the weighted half-line estimate. Lemma 4.4 and Proposition 5.1 now state the precise smoothness, convexity, slope-range, bounded-derivative, and endpoint assumptions; Proposition 4.2 treats $c = 1$ explicitly; and the zero-crossing conclusion is stated for interior zeros. Most importantly, Lemma A.2 proves the fixed-parameter endpoint expansion directly from the initial expansion, including the mixed time-slope remainder bounds, so it no longer invokes Lemma 3.2 outside that lemma's Parisi-profile setting. Its proof first obtains spatial remainder bounds through order eight before deriving one time derivative through order six. Proposition A.3 also records the exact strips and lateral boundary data used in its three maximum-principle applications. The corrected logarithmic derivative (52) agrees with equation (4.90) of [3]; all subsequent identities (66)–(72) require, and are consistent with, that sign. No theorem statement or central algebraic identity in Sections 4–7 is changed by these repairs. The genuinely new steps remain Lemma 3.2, Proposition 4.2, and the application of Proposition 5.1 at the origin. The theorem proves an interval beginning at zero for every $\beta > 1$; it does not assert that the full support is one interval or exclude gaps or atoms away from the initial interval.

A Fixed-parameter preservation of the slope cone

We record the invariant calculation underlying Lemma 4.4. Let

$$u_r = \frac{1}{2}(u_{xx} + au_x^2), \quad a > 0, \quad (88)$$

with an even C^∞ initial profile ψ satisfying $\psi'(\mathbb{R}) = (-1, 1)$, $\psi'' > 0$, bounded positive-order derivatives, and the differentiated endpoint expansion (12). At fixed r , set

$$B = u_x, \quad C = u_{xx},$$

and use B as the spatial coordinate. Define K, J by (25). All r -derivatives below are taken at fixed B .

Lemma A.1 (Degenerate strip maximum principle). *Let v be continuous on $[0, R] \times [\alpha_0, \gamma_0]$, smooth in the open strip, and suppose*

$$v_r = dv_{BB} + bv_B + cv + f,$$

where $d > 0$ in the open strip, $d \geq 0$ on the closure, c is bounded above, and $f \geq 0$. If the initial and lateral boundary data are nonnegative, then $v \geq 0$. The reversed statement holds when $f \leq 0$ and all data are nonpositive.

Proof. Choose $\lambda > \sup c$ and set $\tilde{v} = e^{-\lambda r}v$. If $\tilde{v}$ had a negative minimum, the boundary assumptions would force it to occur at a point with $r > 0$ and $B \in (\alpha_0, \gamma_0)$. There, using the left time derivative if $r = R$,

$$\tilde{v}_r \leq 0, \quad \tilde{v}_B = 0, \quad \tilde{v}_{BB} \geq 0.$$

The PDE gives

$$\tilde{v}_r = d\tilde{v}_{BB} + b\tilde{v}_B + (c - \lambda)\tilde{v} + e^{-\lambda r}f > 0,$$

a contradiction. Degeneracy of d at the lateral boundary is harmless because a negative minimum cannot occur there. Apply the result to $-v$ for the reversed statement. $\square$

Differentiating $B = u_x(r, x(r, B))$ and $C = u_{xx}(r, x(r, B))$ at fixed B gives

$$x_r = BK, \quad C_r = -C^2 J. \quad (89)$$

Twice differentiating the second equation in B yields

$$J_r = \frac{C^2}{2} J_{BB} + 2CC_B J_B + (C_B^2 - 2C(J+a))J. \quad (90)$$

For $\kappa = J_B$, one further differentiation gives

$$\begin{aligned} \kappa_r = \frac{C^2}{2} \kappa_{BB} + 3CC_B \kappa_B + (3C_B^2 - 6C(J+a) - 2CJ)\kappa \\ - 6C_B(J+a)J. \end{aligned} \quad (91)$$

Set

$$E = C^{3/2}J. \quad (92)$$

Direct substitution gives

$$E_r = \frac{C^2}{2} E_{BB} + \frac{CC_B}{2} E_B - VE - \Lambda E^2, \quad (93)$$

where

$$V = \frac{C_B^2}{8} + \frac{C(J+a)}{2}, \quad \Lambda = \frac{3}{2}C^{-1/2}. \quad (94)$$

Since $C_{BB} = -2(J+a)$,

$$V_B = \frac{C}{2}J_B, \quad \Lambda_B = -\frac{3}{4}C^{-3/2}C_B. \quad (95)$$

Thus $F = E_B$ satisfies

$$\begin{aligned} F_r = \frac{C^2}{2} F_{BB} + \frac{3CC_B}{2} F_B + \left[\left(\frac{CC_B}{2} \right)_B - V - 2\Lambda E \right] F \\ - V_B E - \Lambda_B E^2. \end{aligned} \quad (96)$$

Lemma A.2 (Uniform mixed endpoint expansion). *Fix $R_0 > 0$. Under the fixed-parameter flow (88), there are functions $c_2, c_3 \in C^1([0, R_0])$ and a remainder $\mathcal{R}$ such that, with $\delta = 1 - B$,*

$$C(r, B) = 2\delta + c_2(r)\delta^2 + c_3(r)\delta^3 + \mathcal{R}(r, B), \quad (97)$$

and, uniformly for $0 \leq r \leq R_0$ and B sufficiently close to 1,

$$|\partial_r^\alpha \partial_B^\ell \mathcal{R}(r, B)| \leq C_{\alpha, \ell, R_0} \delta^{4-\ell}, \quad \alpha \in \{0, 1\}, \quad 0 \leq \ell \leq 3. \quad (98)$$

In particular, C, J, J_B extend continuously to $B = 1$; after simplifying the products involving $C^{-1/2}$, every coefficient and source term appearing in (90)–(96) has the boundary regularity required by Lemma A.1. Moreover,

$$(C^{3/2}J)_B(r, 1) = 0. \quad (99)$$

Proof. We establish (100)–(101) directly for the fixed-parameter flow, using Lemma 3.2 only to supply the endpoint expansion of the initial profile $\psi = u(0, \cdot)$. Write $y = e^{-2x}$. Let $c(0), d_1(0), d_2(0), d_3(0)$ be its endpoint coefficients, with $d_1(0) > 0$, and let c, d_1, d_2, d_3 solve (15)–(18) with the constant value a . Then, uniformly for $0 \leq r \leq R_0$,

$$u(r, x) = x + c(r) + d_1(r)y + d_2(r)y^2 + d_3(r)y^3 + \mathcal{E}(r, x), \quad (100)$$

where $c, d_1, d_2, d_3 \in C^1([0, R_0])$, d_1 is bounded away from zero, and

$$|\partial_r^\alpha \partial_x^k \mathcal{E}(r, x)| \leq C_{\alpha, k, R_0} e^{-8x}, \quad \alpha \in \{0, 1\}, \quad 0 \leq k \leq 6. \quad (101)$$

Indeed, let $P = x + c + d_1 y + d_2 y^2 + d_3 y^3$ and $E = u - P$. Matching the first four endpoint powers gives

$$E_r = \frac{1}{2} E_{xx} + \frac{a}{2} (u_x + P_x) E_x - F, \quad \partial_x^k F = O_k(e^{-8x}).$$

At $r = 0$, the assumed differentiated expansion of ψ gives $\partial_x^k E(0, x) = O_k(e^{-8x})$. The Cole–Hopf representation and the bounded-derivative hypotheses give, on every finite r -interval, linear growth of u , uniform bounds for all positive spatial derivatives of u , and uniform boundary values at $x = X$; the same properties are immediate for P . Lemma 3.1, applied to the displayed equation after differentiating in x through order eight, therefore yields

$$|\partial_x^k E(r, x)| \leq C_{k, R_0} e^{-8x}, \quad 0 \leq k \leq 8.$$

Moreover $d_1(r) = d_1(0)e^{2(1-a)r}$, so d_1 stays bounded away from zero on $[0, R_0]$. Finally, differentiating the displayed evolution up to six times in x and using these order-eight spatial estimates gives the same bounds for $\partial_r \partial_x^k E$, $0 \leq k \leq 6$. This proves (100)–(101) without applying Lemma 3.2 to an arbitrary flow profile. Set $\delta = 1 - B = 1 - u_x(r, x)$.

Differentiating (100) gives

$$\delta = 2d_1 y + 4d_2 y^2 + 6d_3 y^3 + T_1(r, y), \quad (102)$$

$$C = 4d_1 y + 16d_2 y^2 + 36d_3 y^3 + T_2(r, y), \quad (103)$$

where, after viewing the remainders as functions of (r, y) ,

$$|\partial_r^\alpha \partial_y^\ell T_i(r, y)| \leq C_{\alpha, \ell, R_0} y^{4-\ell}, \quad \alpha \in \{0, 1\}, \quad 0 \leq \ell \leq 4, \quad (104)$$

for $i = 1, 2$. To see the y -derivative bounds directly, use

$$y^\ell \partial_y^\ell = \left(-\frac{1}{2}\right)^\ell \prod_{j=0}^{\ell-1} (\partial_x + 2j)$$

on a function of $y = e^{-2x}$ and then apply (101).

Let

$$\mathcal{F}(r, y) = 2d_1(r)y + 4d_2(r)y^2 + 6d_3(r)y^3 + T_1(r, y).$$

Uniform positivity of d_1 implies that, after reducing $y_0 > 0$ if necessary,

$$\partial_y \mathcal{F}(r, y) \geq c_0 > 0 \quad (0 \leq r \leq R_0, \quad 0 \leq y \leq y_0). \quad (105)$$

Hence $\delta = \mathcal{F}(r, y)$ has a unique inverse $y = Y(r, \delta)$ for small δ , and the parameter-dependent implicit-function theorem gives bounded mixed derivatives $\partial_r^\alpha \partial_\delta^\ell Y$ for $\alpha \leq 1$ and $\ell \leq 4$.

Choose $b_1, b_2, b_3 \in C^1([0, R_0])$ so that

$$Y_0(r, \delta) = b_1(r)\delta + b_2(r)\delta^2 + b_3(r)\delta^3$$

cancels the coefficients of $\delta, \delta^2, \delta^3$ in $\mathcal{F}(r, Y_0) - \delta$. The bounds (104) then imply

$$|\partial_r^\alpha \partial_\delta^\ell (\mathcal{F}(r, Y_0(r, \delta)) - \delta)| \leq C_{\alpha, \ell, R_0} \delta^{4-\ell}, \quad \alpha \in \{0, 1\}, \quad 0 \leq \ell \leq 3. \quad (106)$$

By the mean-value identity,

$$A(r, \delta)(Y(r, \delta) - Y_0(r, \delta)) = -(\mathcal{F}(r, Y_0(r, \delta)) - \delta), \quad (107)$$

where

$$A(r, \delta) = \int_0^1 \partial_y \mathcal{F}(r, Y_0 + \theta(Y - Y_0)) d\theta.$$

By (105), $A \geq c_0$, and the mixed derivatives of A through the required orders are bounded. Differentiating (107) inductively in r and δ , using (106), yields

$$|\partial_r^\alpha \partial_\delta^\ell (Y - Y_0)| \leq C_{\alpha,\ell,R_0} \delta^{4-\ell}, \quad \alpha \in \{0,1\}, \quad 0 \leq \ell \leq 3. \quad (108)$$

Substitute $y = Y(r, \delta)$ into (103). Equations (104) and (108) give (97)–(98); the linear coefficient is $4d_1/(2d_1) = 2$.

Finally,

$$J = -a - \frac{1}{2}C_{BB}, \quad J_B = -\frac{1}{2}C_{BBB}.$$

Thus (98) gives the joint continuity and endpoint traces of J, J_B . Since $C = O(\delta)$, $C_B = O(1)$, $J = O(1)$, and $J_B = O(1)$,

$$(C^{3/2}J)_B = \frac{3}{2}C^{1/2}C_B J + C^{3/2}J_B = O(\delta^{1/2}) \rightarrow 0,$$

which proves (99). The combinations containing $C^{-1/2}$ in (93)–(96) reduce to the bounded expressions displayed below, while every other coefficient follows directly from (98). $\square$

Substitution of (97) into $C_r = -C^2 J$ is now legitimate coefficient by coefficient and gives

$$c'_2 = 4(a + c_2), \quad c'_3 = 12c_3 + 4c_2(a + c_2). \quad (109)$$

Hence the endpoint values

$$j_\infty = J(r, 1) = -a - c_2(r), \quad \kappa_\infty = J_B(r, 1) = 3c_3(r) \quad (110)$$

satisfy

$$j'_\infty = 4j_\infty, \quad \kappa'_\infty = 12\kappa_\infty + 12(j_\infty + a)j_\infty. \quad (111)$$

Thus nonnegative endpoint data remain nonnegative. Lemma A.2, together with smoothness on compact subsets of $(-1, 1)$, verifies every boundary trace and coefficient regularity required by the maximum-principle equations below.

Proposition A.3 (Fixed-parameter preservation). *If the five cone inequalities (27) hold at $r = 0$, then they hold for every $r \geq 0$ under the flow (88).*

Proof. The functions C and J are even in B . Apply Lemma A.1 to (90) on the strip $[0, R] \times [-1, 1]$. The later applications to $\kappa = J_B$ and $F = (C^{3/2}J)_B$ are on $[0, R] \times [0, 1]$, with $B = 0$ treated as a lateral boundary; evenness gives $\kappa(r, 0) = F(r, 0) = 0$. Since the endpoint values of J are $j_\infty(r) \geq 0$ by (111), the maximum principle gives $J \geq 0$. Since $J = (BK)_B$ and $BK(0) = 0$,

$$BK(B) = \int_0^B J(v) dv, \quad (112)$$

so $K \geq 0$ and $C_B = -2(a + K)B \leq 0$ for $B \geq 0$.

In (91), the source $-6C_B(J + a)J$ is now nonnegative. The boundary values are $\kappa(r, 0) = 0$ by evenness and $\kappa(r, 1) = \kappa_\infty(r) \geq 0$ by (111). The maximum principle gives $J_B = \kappa \geq 0$. Thus J is nondecreasing, and (112) gives

$$K(B) = \frac{1}{B} \int_0^B J(v) dv \leq J(B).$$

Since $J = K + BK_B$, this yields $K_B \geq 0$.

Finally, $E \geq 0$, and (95) gives $V_B \geq 0$ and $\Lambda_B \geq 0$. The last two terms in (96) are therefore nonpositive. The apparently singular combinations extend continuously to $B = 1$ because

$$\Lambda E = \frac{3}{2}CJ, \quad \Lambda_B E^2 = -\frac{3}{4}C^{3/2}C_B J^2.$$

At $B = 0$, evenness gives $F(r, 0) = 0$; at $B = 1$, Lemma A.2 gives $F(r, 1) = 0$. The reversed maximum principle yields

$$F = (C^{3/2}J)_B \leq 0.$$

This proves preservation of all five inequalities. $\square$

If the parameter is decreased from a to $b \in [0, a]$ while u is fixed, then

$$K_b = K_a + a - b, \quad J_b = J_a + a - b, \quad (J_b)_B = (J_a)_B, \quad (113)$$

and

$$(C^{3/2}J_b)_B = (C^{3/2}J_a)_B + \frac{3}{2}(a-b)C^{1/2}C_B \leq (C^{3/2}J_a)_B. \quad (114)$$

Thus parameter decrease also preserves the cone, including the limiting value $b = 0$.

Corollary A.4 (Finite-profile cone). *Begin with $(u, a) = (\log \cosh, 1)$ and perform finitely many operations of the following forms:*

(1) *replace u by $T_{a,r}u$ for $r \geq 0$, leaving a fixed;*

(2) *replace a by $b \in [0, a]$, leaving u fixed.*

Every resulting pair lies in the closed slope cone.

Proof. For $u = \log \cosh x$, one has

$$B = \tanh x, \quad C = 1 - B^2.$$

At parameter $a = 1$,

$$K = -\frac{-2B}{2B} - 1 = 0, \quad J = -1 - \frac{1}{2}(-2) = 0.$$

Thus all five cone inequalities hold initially. Proposition A.3 and (113)–(114) preserve them under every permitted operation. $\square$

References

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Appendix R

Expanded technical notes in normal type

Companion to *From Infinite Support to an Initial Full-RSB Interval*

Purpose

This appendix restates, at normal body size, the analytic-completion material that was compressed into small-font notes in the fully fixed manuscript. It does not replace the numbered proofs. It records the exact approximation choices, endpoint bounds, derivative budgets, and maximum-principle boundary data on which those proofs rely.

R.1 Revision status, sign correction, and scope

The sign-critical logarithmic derivative in the slope coordinate is

$$\frac{w_B}{w} = -\frac{N+z}{C}.$$

Indeed, from the one-time density and the definition $w = 2p_s(x(B))C(B)$,

$$\frac{w_B}{w} = \frac{mB - x/s}{C} + \frac{C_B}{C} = \frac{mB - x/s - 2z}{C} = -\frac{N+z}{C},$$

because $z = mB + P$ and $N = x/s + P$. The subsequent identities in the zero-crossing proof are consistent with this sign.

The final revision incorporates the following analytic repairs without changing the theorem statement or the sign-sensitive algebra in Sections 4–7:

- Lemma 2.3 constructs atomic approximations that remain exactly flat on a prescribed constant-profile interval.
- Lemma 3.1 uses an escape barrier whose degree dominates the assumed polynomial growth and requires the coefficient derivatives generated by differentiation.
- Lemma 3.2 verifies all boundary traces at the finite endpoint $x = X$.
- Proposition 4.2 treats both endpoint cases $c = 0$ and $c = 1$.
- Lemma 4.4 and Proposition 5.1 state the full smoothness, convexity, slope-range, bounded-derivative, and endpoint hypotheses.
- Proposition 5.1 concerns interior zeros of q'' .
- Lemma A.2 proves the fixed-parameter endpoint expansion directly from the initial expansion rather than applying Lemma 3.2 outside its stated Parisi-profile class.
- Proposition A.3 identifies the precise strips and lateral boundary values used in the three maximum-principle applications.

Exact novelty and exact scope

The central new steps are Lemma 3.2, Proposition 4.2, and the application of Proposition 5.1 at the origin. The theorem produces an interval beginning at zero for every $\beta > 1$. It does not claim that the entire support is one interval, nor does it rule out gaps or atoms away from the initial interval.

R.2 Flat-segment atomic approximation and higher Itô identities

Let $I = [r_0, r_1]$ be compactly contained in an open interval on which the distribution function α_ν is constant, say $\alpha_\nu \equiv m$. Choose

$$a < r_0 < r_1 < b$$

inside that interval. Then

$$\nu((a, b]) = 0, \quad \nu([0, a]) = m, \quad \nu((b, 1]) = 1 - m.$$

Approximate the finite measures $\nu|_{[0, a]}$ and $\nu|_{(b, 1]}$ separately by finitely atomic measures with exactly the same total masses. The zero measure is allowed when $m = 0$ or $m = 1$. Let ν_n be the sum of the two approximations. Then

$$\nu_n \rightarrow \nu \quad \text{in the Parisi profile metric,} \quad \alpha_n \equiv m \quad \text{on } [a, b],$$

so the approximants retain the same flat segment on a neighborhood of I .

For each atomic profile, differentiating the PDE three times and applying Itô's formula along the corresponding optimal diffusion gives

$$du_{n,xxx}(s, X_s^n) = -3\beta^2 m u_{n,xx}(s, X_s^n) u_{n,xxx}(s, X_s^n) ds + \beta u_{n,xxx}(s, X_s^n) dW_s.$$

Uniform convergence of the required spatial derivatives, convergence of the coupled diffusions, dominated convergence for the drift, and Itô isometry for the stochastic integral pass this identity to the limit on I . Repeating the differentiated calculation gives every higher identity used on a support gap.

The same approximation argument supplies the arbitrary-profile identities

$$dM_s = \beta C_s dW_s, \quad dC_s = -\beta^2 \alpha_\nu(s) C_s^2 ds + \beta D_s dW_s,$$

with

$$\Gamma'_\nu(s) = \beta^2 \mathbb{E}[C_s^2]$$

and

$$\Gamma'_\nu(s) - \Gamma'_\nu(0) = \beta^4 \int_0^s \mathbb{E}[D_r^2 - 2\alpha_\nu(r) C_r^3] dr.$$

Bounded spatial derivatives ensure that all stochastic integrals appearing here are true martingales.

R.3 Weighted half-line estimate and arbitrary-profile endpoint expansion

For the reversed Parisi flow, subtract the truncated expansion

$$P(r, x) = x + c(r) + d_1(r)e^{-2x} + d_2(r)e^{-4x} + d_3(r)e^{-6x}$$

from the PDE solution. The coefficient ODEs cancel all terms through e^{-6x} , leaving a forcing of order e^{-8x} after every required spatial derivative.

The weighted maximum-principle argument on $[X, \infty)$ has two separate barriers:

- an exponentially decaying barrier $Ae^{\gamma r}e^{-\lambda(x-X)}$ to dominate the initial data, the boundary data at $x = X$, and the forcing;
- an escape barrier of degree strictly larger than the assumed polynomial growth, for example

$$\eta e^{\rho r}(1 + (x - X)^{2M+2}),$$

which permits comparison first on $[X, Y]$, then $Y \rightarrow \infty$, and finally $\eta \downarrow 0$.

After differentiating j times, the equation contains derivatives of the coefficients through order j ; these are included among the hypotheses. The same global barrier works across finitely many time slabs because the differentiated remainder is continuous at the interfaces.

The boundary values at the finite endpoint are also uniform. The Cole–Hopf representation gives, for fixed X and a bounded time interval,

$$\sup_r |v(r, X)| \leq |X| + C,$$

while the all-order spatial derivative bounds control every positive-order derivative of v ; the polynomial ansatz P and its derivatives are uniformly bounded at X . Consequently every boundary trace required by the half-line estimate is controlled.

This yields, for an arbitrary Parisi profile,

$$\psi(x) = x + c_0 + d_1 e^{-2x} + d_2 e^{-4x} + d_3 e^{-6x} + O(e^{-8x}), \quad d_1 > 0,$$

with the differentiated remainder bounds required to pass to the slope variable. Setting $B = \psi'(x)$, $C(B) = \psi''(x(B))$, and $\delta = 1 - B$, one obtains

$$C(B) = 2\delta + c_2 \delta^2 + c_3 \delta^3 + O(\delta^4),$$

with three controlled B -derivatives.

R.4 Closure of the slope cone and endpoint cases

For an arbitrary probability measure ν , a time $q \in [0, 1]$, and

$$c = \nu([0, q]), \quad \psi = u_\nu(q, \cdot),$$

collapse all mass below q to an atom at q . Only the profile above q affects $u_\nu(q, \cdot)$. Approximate the remaining measure by finite atomic measures. The finite-profile cone then applies, and all slope quantities converge uniformly on compact slope intervals because the spatial derivatives converge uniformly and the inverse slope maps converge.

The endpoint cases are explicit:

- If $c = 1$, there is no mass above q ; take the approximating measure above q to be zero.
- If $c = 0$, introduce a small atom $\varepsilon_n \delta_q$, apply the finite-profile cone at parameter ε_n , decrease the parameter to zero, and then let $\varepsilon_n \downarrow 0$.

There is no singularity at $B = 0$. Evenness gives $C_B(0) = 0$, and the identities

$$\frac{C_B(B)}{B} = \int_0^1 C_{BB}(\tau B) d\tau$$

and

$$\frac{BC_{BB}(B) - C_B(B)}{B^2} = \frac{1}{B^2} \int_0^B \int_v^B C_{BBB}(w) dw dv$$

control the apparently singular expressions uniformly through the origin.

The passage to the limit preserves only the *closed* cone inequalities. The strict inequality needed later is not inferred by limit; Section 5 proves it directly for the limiting profile.

R.5 Exact abstract hypotheses and preservation of strict convexity

The semigroup and generalized gap arguments use the following hypotheses on the initial profile ψ :

- $\psi \in C^\infty(\mathbb{R})$ is even and strictly convex;
- $\psi'(\mathbb{R}) = (-1, 1)$;
- every positive-order spatial derivative is bounded;
- the differentiated endpoint expansion of Lemma 3.2 holds;
- (ψ, a) lies in the closed slope cone for the relevant parameter $a > 0$.

Under the tilted Gaussian law associated with the Cole–Hopf transform,

$$(T_{a,r}\psi)_{xx}(x) = \mathbb{E}_{Q_{r,x}}[\psi''(x + \sqrt{r}Z)] + a \operatorname{Var}_{Q_{r,x}}(\psi'(x + \sqrt{r}Z)) > 0.$$

Thus strict convexity, the global slope coordinate, and the slope range persist under the flow.

For the generalized gap inequality, take $m, t > 0$ and

$$U(s, x) = T_{m,t-s}\psi(x), \quad dX_s = mU_x(s, X_s) ds + d\widetilde{W}_s,$$

with

$$q(s) = \mathbb{E}[U_x(s, X_s)^2], \quad \theta = U_{xx}(0, 0) > 0.$$

The conclusion is

$$2 \int_0^t q(s) \, ds = tq(t) \implies q(t) < t\theta^2.$$

All references to zeros of q'' in this argument concern zeros in the open interval $(0, t)$.

R.6 Differentiation under the slope integral and the one-crossing argument

In slope coordinates,

$$\frac{1}{2}q''(s) = \int_0^1 w(s, B)\Phi(s, B) \, dB.$$

The endpoint expansion gives

$$x(B) = \frac{1}{2} \log \frac{1}{1-B} + O(1), \quad C(B) = 2(1-B) + O((1-B)^2).$$

Since U has linear growth, the density contains the Gaussian factor $\exp(-x^2/(2s))$. On every compact interval $I \Subset (0, t)$, one may dominate the differentiated integrand near $B = 1$ by a function of the form

$$C_I(1-B)(1+L(B)^2) \exp(-c_I L(B)^2 + C_I L(B)), \quad L(B) = \log \frac{1}{1-B},$$

which is integrable and independent of $s \in I$. This justifies differentiation under the integral and the integrations by parts used in the zero-crossing lemma.

At a zero of q'' in $(0, t)$, the slope-cone inequalities imply

$$q'''(s) > 0.$$

Thus every interior zero is crossed upward. Since

$$q''(0) = -2m\theta^3 < 0,$$

there can be at most one interior zero: after an upward crossing, a later first return to zero would have nonpositive left derivative, contradicting the strict positive derivative at that zero.

The balance identity

$$tq(t) - 2 \int_0^t q(s) \, ds = \int_0^t s(t-s)q''(s) \, ds$$

then forces a unique sign change from negative to positive. Comparing the first moments of the positive and negative parts shows

$$q(t) - tq'(0) < 0,$$

and $q'(0) = \theta^2$ gives the strict generalized gap inequality.

R.7 Direct fixed-flow endpoint expansion and derivative budget

For the fixed-parameter flow

$$u_r = \frac{1}{2}(u_{xx} + au_x^2), \quad a > 0,$$

Lemma A.2 is proved directly from the endpoint expansion of the initial profile. Writing $y = e^{-2x}$,

$$u(r, x) = x + c(r) + d_1(r)y + d_2(r)y^2 + d_3(r)y^3 + E(r, x),$$

where $d_1(r) = d_1(0)e^{2(1-a)r}$ remains bounded away from zero on compact time intervals.

The derivative bookkeeping is as follows:

1. Apply the weighted half-line estimate after differentiating the remainder equation in x through order eight:

$$|\partial_x^k E(r, x)| \leq Ce^{-8x}, \quad 0 \leq k \leq 8.$$

2. Differentiate the evolution once in time and up to six times in x . The order-eight spatial estimates control the two additional derivatives produced by the second-order PDE, giving

$$|\partial_r \partial_x^k E(r, x)| \leq Ce^{-8x}, \quad 0 \leq k \leq 6.$$

3. Invert $\delta = 1 - u_x(r, x)$ as a function of y . Pure δ -derivatives are needed through order four; mixed derivatives with one r -derivative are needed through order three. This is precisely the budget required for C , J , J_B , and the boundary traces in the maximum-principle equations.

The resulting expansion is

$$C(r, B) = 2\delta + c_2(r)\delta^2 + c_3(r)\delta^3 + \mathcal{R}(r, B), \quad \delta = 1 - B,$$

with

$$|\partial_r^\alpha \partial_B^\ell \mathcal{R}(r, B)| \leq C\delta^{4-\ell}, \quad \alpha \in \{0, 1\}, \quad 0 \leq \ell \leq 3.$$

In particular,

$$(C^{3/2}J)_B(r, 1) = 0.$$

Substitution into $C_r = -C^2J$ gives the endpoint ODEs

$$j'_\infty = 4j_\infty, \quad \kappa'_\infty = 12\kappa_\infty + 12(j_\infty + a)j_\infty.$$

The initial cone signs and continuous endpoint extension give

$$j_\infty(0) \geq 0, \quad \kappa_\infty(0) \geq 0.$$

The ODEs therefore preserve nonnegativity for all later times.

R.8 Maximum-principle strips and lateral boundary data

The coefficients in the degenerate parabolic equations are continuous on the relevant closed strips after the endpoint combinations containing $C^{-1/2}$ are simplified; on compact time intervals they are locally bounded, and the zero-order coefficients are bounded above. The three applications are:

1. Apply the maximum principle to J on

$$[0, R] \times [-1, 1].$$

The endpoint values at $B = \pm 1$ follow from the endpoint ODEs and evenness.

2. Apply it to $\kappa = J_B$ on

$$[0, R] \times [0, 1].$$

Treat $B = 0$ as a lateral boundary. Evenness gives $\kappa(r, 0) = 0$, and the endpoint expansion gives $\kappa(r, 1) = \kappa_\infty(r) \geq 0$.

3. Apply the reversed maximum principle to

$$F = (C^{3/2}J)_B$$

on $[0, R] \times [0, 1]$. Evenness gives $F(r, 0) = 0$, and the fixed-flow endpoint expansion gives $F(r, 1) = 0$.

These boundary data yield

$$J \geq 0, \quad J_B \geq 0, \quad (C^{3/2}J)_B \leq 0,$$

and then the integral identity $BK(B) = \int_0^B J(v) dv$ gives $K \geq 0$ and $K_B \geq 0$.

R.9 Uniformity at small support gaps

After exact marginality and the absence of an atom at zero have been proved,

$$C_s \rightarrow \frac{1}{\beta}, \quad D_s \rightarrow 0, \quad R_s \rightarrow \kappa = u_{xxxx}(0,0) \neq 0$$

in every finite L^p as $s \downarrow 0$, while the gap parameter $m = \alpha(s)$ tends to zero. On a support gap,

$$\Gamma'''(s) = \beta^6 \mathbb{E}[R_s^2 - 12mC_sD_s^2 + 6m^2C_s^4].$$

The bracket converges to $\kappa^2 > 0$ along any sequence of gap points tending to zero. If no uniform ε_β existed, one could choose gap points $s_n \downarrow 0$ where the bracket is nonpositive, contradicting that convergence. Hence $\Gamma'''(s) > 0$ on every sufficiently small support gap.

Therefore Γ' is strictly convex on such a gap (a, b) . Since the endpoints lie in the support,

$$\frac{1}{b-a} \int_a^b \Gamma'(s) \, ds = \frac{\Gamma(b) - \Gamma(a)}{b-a} = 1,$$

while stability gives $\Gamma'(a), \Gamma'(b) \leq 1$. Strict Hermite–Hadamard would imply

$$1 < \frac{\Gamma'(a) + \Gamma'(b)}{2} \leq 1,$$

a contradiction. Thus no sufficiently small complementary interval exists, and the support contains an initial interval.

Audit status

After the repairs summarized above, the manuscript is internally complete relative to the standard Parisi-PDE regularity and optimizer results it quotes and the finite-profile slope calculation reproduced in Appendix A. This is an internal proof audit, not a substitute for independent specialist peer review.